# Northstar Operations Room

WYRESTORM | TECHNICAL PROPOSAL

## Document control

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WYR-WWM800
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## Executive Summary

Route two operator sources to four displays with resilient 4K distribution.

This document explains what we recommend, how it will work day to day, the equipment involved, and what still needs confirming before it becomes a final proposal.

## Room at a glance

### Operations room

Route two operator sources to four displays with resilient 4K distribution.

#### Room and design details

Room dimensions, capacity and site conditions require confirmation.

#### What this space delivers

**Operational continuity:** Operators retain controlled access to approved sources.

#### In this room

Items are grouped by the job each one does in the room, and the number matches the equipment schedule further on. Anything marked optional is still your decision, and work by others is listed under Complete System Design and Responsibilities.

**Encode operator sources**

| ITEM | MODEL | DESCRIPTION | QTY | STATUS |
|---|---|---|---|---|
| 1 | NHD-600-E-TX | NetworkHD 600 encoder | 2 | Required |

**Decode display feeds**

| ITEM | MODEL | DESCRIPTION | QTY | STATUS |
|---|---|---|---|---|
| 2 | NHD-600-E-RX | NetworkHD 600 decoder | 4 | Required |

**Routing control**

| ITEM | MODEL | DESCRIPTION | QTY | STATUS |
|---|---|---|---|---|
| 3 | NHD-CTL-PRO-V2 | NetworkHD controller | 1 | Required |

## Solution Overview

NetworkHD

**What successful delivery looks like**

Every approved source-to-display route passes functional testing.

## What You Asked For

Route two operator sources to four displays with resilient 4K distribution.

## Discovery Conversation

Below is the conversation this proposal is based on: the question we asked, the answer we recorded, and your own words where we captured them. "Confirmed with customer" means we checked that point with you; "To be confirmed" means it is still open and will be settled before the design is signed off.

| QUESTION ASKED | ANSWER WE RECORDED | CUSTOMER WORDING | STATUS | CAPTURE CONFIDENCE |
|---|---|---|---|---|
| How many displays must receive independently routed content? | Four displays | Each operator needs independent routing. | Confirmed with customer | High confidence (12) |

"Capture confidence" shows how closely your own words matched the answer we recorded. High confidence is a close match. Matched is a close match to a standard phrase. Low confidence — verify before quote is a partial match we will check with you before quoting. The number in brackets is the match score.

If a row has a note but no clear answer, that point was still open when this was written and will be confirmed before the design is signed off.

## Recommended Solution

**Recommendation: NetworkHD** is the best fit for this requirement on the evidence gathered so far. Confidence: 96 out of 100.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

**Still to confirm:** redundant power and switch failover behaviour during commissioning.

**Required WyreStorm products:** NHD-600-E-TX - NetworkHD 600 encoder; NHD-600-E-RX - NetworkHD 600 decoder; NHD-CTL-PRO-V2 - NetworkHD controller

## Practical Operation

This is what the system will do for the people using the room, day to day.

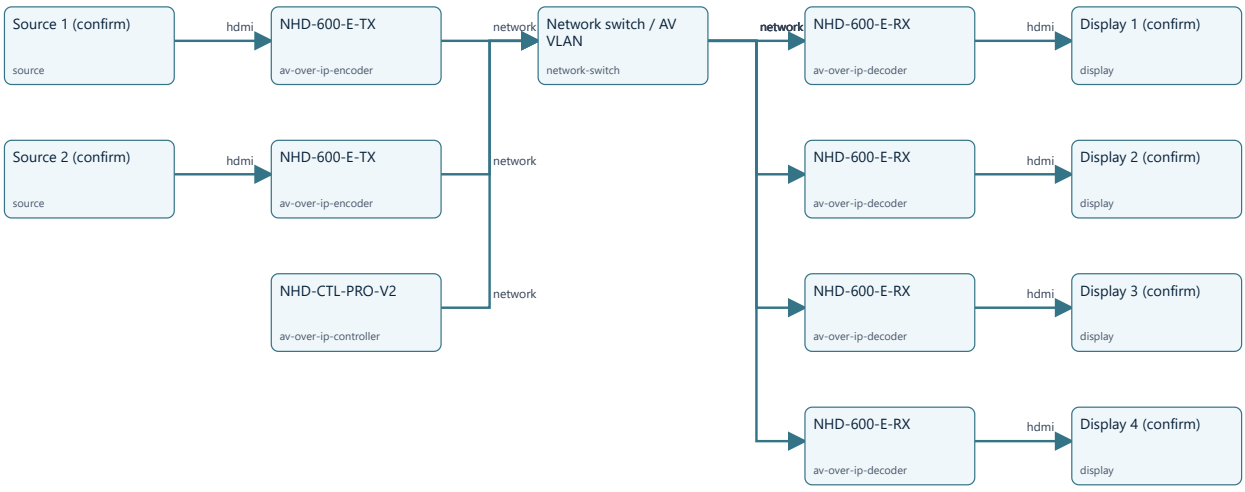
Operators select an approved source and route it to any display.

## Commercial and Operational Benefits

A consistent, repeatable connection experience across the room(s) covered by this proposal.
AV-over-IP provides a potential expansion path, subject to compatible endpoints, network capacity and the final design.
We will tie these benefits to your own business measures before this proposal is issued.

## Technical Architecture

The diagram and cable schedule below show how the system connects together. Both are built from the equipment schedule further on.




Concept drawing: confirm physical connections and site interfaces before installation.

Transport is shown generically because signal distance or connector evidence is not confirmed. Connections without a verified interface profile are proposed topology only. Confirm named ports, endpoint pairing, power and cable types before installation.

Control routing requires design: Specify the controller and controlled-device interfaces before allocating control cables.
Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
Cable length not confirmed for hdmi run: The Decoder to display connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
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Power source unconfirmed for NHD-600-E-TX: "NetworkHD 600 encoder" (NHD-600-E-TX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
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Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-CTL-PRO-V2: "NetworkHD controller" (NHD-CTL-PRO-V2) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.

This diagram shows the signal path we expect, from source to screen, so you and the installer are looking at the same picture. The cable schedule below gives the cable type, connectors, maximum length and what still needs checking for each connection.

| CABLE RUN | TRANSPORT | MAX LENGTH | CONNECTORS | STATUS | VALIDATION REMINDER |
|---|---|---|---|---|---|
| Source 1 (confirm) → NHD-600-E-TX | Video | 15m | SOURCE PORT TBD to HDMI IN | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |
| Source 2 (confirm) → NHD-600-E-TX | Video | 15m | SOURCE PORT TBD to HDMI IN | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |
| NHD-600-E-TX → Network switch | Network / AVoIP | Not confirmed | AV LAN (confirm) to SWITCH PORT 1 | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| NHD-600-E-TX → Network switch | Network / AVoIP | Not confirmed | AV LAN (confirm) to SWITCH PORT 2 | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| Network switch → NHD-600-E-RX | Network / AVoIP | Not confirmed | SWITCH PORT 3 to AV LAN (confirm) | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |

| CABLE RUN | TRANSPORT | MAX LENGTH | CONNECTORS | STATUS | VALIDATION REMINDER |
|---|---|---|---|---|---|
| Network switch → NHD-600-E-RX | Network / AVoIP | Not confirmed | SWITCH PORT 4 to AV LAN (confirm) | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| Network switch → NHD-600-E-RX | Network / AVoIP | Not confirmed | SWITCH PORT 5 to AV LAN (confirm) | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| Network switch → NHD-600-E-RX | Network / AVoIP | Not confirmed | SWITCH PORT 6 to AV LAN (confirm) | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| NHD-CTL-PRO-V2 → Network switch | Network / AVoIP | Not confirmed | AV LAN (confirm) to SWITCH PORT 7 | Needs site confirmation | Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology. |
| NHD-600-E-RX → Display 1 (confirm) | Video | 15m | HDMI OUT to DISPLAY INPUT (confirm) | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |

| CABLE RUN | TRANSPORT | MAX LENGTH | CONNECTORS | STATUS | VALIDATION REMINDER |
|---|---|---|---|---|---|
| NHD-600-E-RX → Display 2 (confirm) | Video | 15m | HDMI OUT to DISPLAY INPUT (confirm) | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |
| NHD-600-E-RX → Display 3 (confirm) | Video | 15m | HDMI OUT to DISPLAY INPUT (confirm) | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |
| NHD-600-E-RX → Display 4 (confirm) | Video | 15m | HDMI OUT to DISPLAY INPUT (confirm) | Needs site confirmation | Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements. |
| AV network infrastructure | Network / AVoIP | Not confirmed | RJ45 to RJ45 | Needs site confirmation | Confirm switch model, port count, PoE budget, VLAN configuration, multicast policy, and uplink capacity. |

## Products in the Solution

Product photograph unavailable.

**NHD-600-E-TX — NetworkHD 600 encoder**
Role in this design: Encode operator sources

Official product link to be confirmed.

Product photograph unavailable.

**NHD-600-E-RX — NetworkHD 600 decoder**
Role in this design: Decode display feeds

Official product link to be confirmed.

Product photograph unavailable.

**NHD-CTL-PRO-V2 — NetworkHD controller**
Role in this design: Routing control

Official product link to be confirmed.

## Complete System Design and Responsibilities

The items below complete the wider solution. Each one is still open: none of it is equipment or work included in this proposal, and we will agree the owner, detail and commercial treatment with you before it is issued.

Displays | Operator displays — Confirm size and mounting. | Qty 4 | Responsibility: Integrator | Scope status: allowance

## Display Schedule

4 displays in scope, each driven by one feed.

| SCREEN | RESOLUTION | FEEDS | SCALING | NOTES |
|---|---|---|---|---|
| Display 1 | To confirm | 1 | Native 1:1, confirm at survey | Operator displays |
| Display 2 | To confirm | 1 | Native 1:1, confirm at survey | |
| Display 3 | To confirm | 1 | Native 1:1, confirm at survey | |
| Display 4 | To confirm | 1 | Native 1:1, confirm at survey | |

**Basis:** Native resolution, feed count and scaling are confirmed when the display models are recorded.

## Equipment Schedule

| ITEM | SKU | DESCRIPTION | ROLE | QTY | TYPE | NOTES |
|---|---|---|---|---|---|---|
| 1 | NHD-600-E-TX | NetworkHD 600 encoder | Encode operator sources | 2 | Required | Install in the equipment rack. |
| 2 | NHD-600-E-RX | NetworkHD 600 decoder | Decode display feeds | 4 | Required | One decoder per display. |
| 3 | NHD-CTL-PRO-V2 | NetworkHD controller | Routing control | 1 | Required | Confirm firmware compatibility. |
| EQUIPMENT SCHEDULE TOTALS | | | | 7 | 3 line items | |

**Evidence basis:** NHD-600-E-TX: Two confirmed operator sources | NHD-600-E-RX: Four independently routed displays | NHD-CTL-PRO-V2: Central routing and control requirement

## Power Strategy

The power figures below come from WyreStorm product data for each selected product. They are the maximum the published specification states, and should be checked against the current datasheet before ordering.

| SKU | QTY | PER-UNIT MAX | TOTAL MAX | POWER DATA (WYRESTORM) |
|---|---|---|---|---|
| NHD-600-E-TX | 2 | Not proven | — | 802.3at PoE+ |
| NHD-600-E-RX | 4 | Not proven | — | 802.3at PoE+ |
| NHD-CTL-PRO-V2 | 1 | 15.4W | 15.4W | 12V DC \| PoE compliant; Max 15.4W, average 8W |

**Stated maximum consumption:** approximately 15W across 1 product with proven figures. 3 products (NHD-600-E-TX, NHD-600-E-RX, NHD-CTL-PRO-V2) require PoE/PoH — confirm the injector or switch PoE budget covers the total.

## Options

No optional enhancements are currently selected.

## Assumptions and Exclusions

### Assumptions

**Assumption:** Customer IT will provide a dedicated 10GbE AV VLAN.

### Exclusions

Display, projector, LED and mounting hardware supply unless listed as a specific line item above.
Electrical, structural and builder's work.
Network switch procurement, VLAN/multicast configuration and IT commissioning access.
Structured cabling installation and certification.
Third-party control programming beyond WyreStorm-native presets.
Installation, labour and commissioning, unless separately quoted.

## Risks and Dependencies

### Risks and open questions

**Risk:** Confirm redundant power and switch failover behaviour during commissioning.

### Dependencies

**Dependency: NET-SWITCH-10G-MANAGED - Managed 10GbE network switch**: Confirm switch model, port budget, VLAN and multicast configuration. Why: NetworkHD endpoints require managed 10GbE switching.

## Implementation and Next Steps

Approve the network design and complete acceptance testing.

> This draft includes the selected WyreStorm products. We will confirm final specifications, accessories and dependencies before it is issued.

> Competitor products are not included in this proposal or its equipment schedule.

### Important Notice & Best-Efforts Disclaimer

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## Confirmed Requirement

Route two operator sources to four displays with resilient 4K distribution.

## Design Assumptions

Customer IT will provide a dedicated 10GbE AV VLAN.

## Recommended Architecture

**NetworkHD** is our recommended product family.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

## Required WyreStorm Products

**NHD-600-E-TX** - NetworkHD 600 encoder
**NHD-600-E-RX** - NetworkHD 600 decoder
**NHD-CTL-PRO-V2** - NetworkHD controller

## Optional Enhancements

No optional enhancements are currently selected.

## Risks / Needs Validation

Confirm redundant power and switch failover behaviour during commissioning.

## Next Steps

Approve the network design and complete acceptance testing.